

Chapter 1

Methodology and Results Interpretation

1.1 Data and Empirical Strategy

We study the association between domestic monetary conditions and FDI net inflows in an unbalanced panel of 11 ASEAN-oriented economies (Brunei, Cambodia, Indonesia, Lao PDR, Malaysia, Myanmar, Philippines, Singapore, Thailand, Timor-Leste, Viet Nam) over 2000–2023 (max 264 country-year observations). FDI net inflows (% of GDP) is the dependent variable. The liquidity channel is proxied by broad money growth (annual %). The cost-of-capital channel is proxied by three alternative interest rates: the nominal deposit rate, the real interest rate, and the lending rate. Controls include inflation (GDP deflator), trade openness (% of GDP), log GDP per capita, and exchange-rate depreciation (log-difference of LCU/USD). All monetary proxies and FDI are used as observed; controls are interpolated or edge-filled under strict rules.

The preferred specification is a two-way fixed-effects (TWFE) model:

$$FDI_{it} = \beta_1 BM_{it} + \beta_2 IR_{it}^{(k)} + \gamma' \mathbf{X}_{it} + \alpha_i + \lambda_t + \varepsilon_{it}, \quad (1.1)$$

where α_i and λ_t are country and year fixed effects. The Hausman test rejects random effects across all specifications. We report Driscoll–Kraay (1998) standard errors, which are robust to heteroskedasticity, serial correlation, and cross-sectional dependence, making them appropriate for small- N , moderate- T macro panels.

1.2 Main Results

Two key findings emerge. First, the liquidity proxy (broad money growth) is generally positive but statistically imprecise across all headline models (Panel A), ranging from 0.035 (M4, $p > 0.10$) to 0.122 (M3, $p < 0.05$). This pattern is directionally consistent with a liquidity channel but lacks precision. Second, the cost-of-capital proxies do not deliver a stable negative association: the deposit and lending rates are positive and imprecise, while the real interest rate is negative but also imprecise (-0.323 , $p > 0.10$). Within- R^2 values are modest (≤ 0.09), indicating limited within-country explanatory power after absorbing fixed effects.

Table 1.1: Two-way fixed-effects regression results with Driscoll–Kraay standard errors

	Panel A: Headline Models					Panel B: Extended Models			
	M1	M2	M3	M4	M5	M6a	M6b	M7a	M7b
Broad money growth (annual %)	0.0460* (0.0277)	0.0451 (0.0388)	0.1223** (0.0572)	0.0351 (0.0426)	0.0627* (0.0360)	0.0446 (0.0395)	0.0422 (0.0386)	0.0494 (0.0432)	0.0453 (0.0392)
Deposit interest rate (%)		0.0922 (0.1242)	0.1274 (0.1521)				0.0883 (0.1217)		0.0887 (0.1700)
Real interest rate (%)				-0.3225 (0.2370)		0.0836 (0.1128)		0.0743 (0.1380)	
Lending interest rate (%)					0.2500 (0.1738)				
Tourism arrivals (Log)						-0.5474 (0.6736)	-0.5260 (0.6588)		
Human capital index								0.7949 (3.2223)	0.1646 (3.2596)
Trade (% GDP)	-0.0014 (0.0123)	-0.0154 (0.0175)	-0.0170 (0.0198)	-0.0126 (0.0184)	-0.0182 (0.0167)	-0.0134 (0.0188)	-0.0120 (0.0195)	-0.0165 (0.0174)	-0.0153 (0.0178)
Inflation (%)	-0.0195 (0.0576)	0.0017 (0.0441)	-0.0322 (0.0485)	-0.2957 (0.1988)	-0.0200 (0.0396)	0.0785 (0.1274)	0.0012 (0.0447)	0.0695 (0.1534)	0.0017 (0.0443)
Log GDP per capita	0.2059 (1.5284)	0.9642* (0.5560)	1.1131 (0.8672)	-0.9521 (1.2826)	1.8815*** (0.5928)	1.2607** (0.5675)	0.8889 (0.5750)	1.2495** (0.5319)	0.9459 (0.6209)
Exchange-rate depreciation (%)	0.0153 (0.0272)	0.0539 (0.0413)	0.0585 (0.0469)	0.0801* (0.0427)	0.0624 (0.0384)	0.0637* (0.0368)	0.0560 (0.0425)	0.0618* (0.0344)	0.0540 (0.0407)
Observations	209	149	145	164	151	151	149	151	149
Countries	10	7	7	8	7	7	7	7	7
Within R^2	0.0305	0.0655	0.0883	0.0352	0.0535	0.0387	0.0393	0.0476	0.0653
Slope-adj. R^2	0.0066	0.0260	0.0486	-0.0017	0.0141	-0.0084	-0.0084	0.0010	0.0189
Max VIF	5.27	7.18	7.26	9.74	12.62	667.94	582.07	179.50	207.32
Pesaran CD p	0.0765	0.3760	0.6651	0.0539	0.3604	0.5234	0.4032	0.4543	0.3713

The dependent variable is FDI net inflows (% of GDP). All columns use two-way fixed effects with country and year effects and Driscoll–Kraay standard errors (Bartlett kernel). Standard errors in parentheses. *, **, *** denote significance at 10%, 5%, 1% levels. Panel A: headline models. Panel B: robustness models adding tourism arrivals (M6a, M6b) or the human capital index (M7a, M7b) to the M4 and M2 baselines respectively. Samples vary by model due to unbalanced panel coverage.

The extended models (Panel B) add tourism arrivals (M6a, M6b) or the human capital index (M7a, M7b) to the real-rate and deposit-rate baselines respectively. Monetary coefficients remain stable, confirming that the headline imprecision is not driven by omitted structural controls. However, multicollinearity rises sharply in these specifications (max VIF > 170), reinforcing their appendix status.

1.3 Robustness Checks

Common-sample re-estimation confirms that coefficient differences across monetary proxies are not artifacts of sample composition. Restricting M1 to the M2 support reduces the broad-money coefficient from 0.046 to 0.037, and the lagged M3 specification is stable to this restriction. The lending-rate coefficient in M5 is not robust to harmonization with the M2 sample. These checks are reported in the companion robustness table.

Drop-collinearity diagnostics show that the severe VIF in M7a (387.19) is driven by the joint inclusion of the human capital index and log GDP per capita; omitting either variable resolves the issue (max VIF ≤ 16.80). The human-capital coefficient itself is imprecise regardless of specification (e.g., M7b: 0.165, SE 3.260).

1.4 Limitations

Three caveats shape interpretation. (1) Short panel and identification: With at most 24 annual observations per country before missingness, and as few as 145 post-missingness, year fixed effects absorb common shocks and limited within-country variation remains for monetary variables. Coefficients are conditional associations, not causal effects. (2) Unbalanced proxy coverage: Cambodia has zero observed interest-rate years; Lao PDR and Timor-Leste have partial coverage. Estimation samples vary across models, and results may partly reflect the specific country composition of each specification. (3) Diagnostic fragility: Within- R^2 values are near zero; the real-rate model (M4) shows marginal cross-sectional dependence ($p = 0.054$); the extended models (M6–M7) have extreme multicollinearity. The most defensible conclusion is that monetary conditions show directionally plausible but statistically fragile associations with FDI inflows in this panel.

Appendix

1.4.1 Summary Statistics

Table 1.2 presents descriptive statistics for the processed panel. FDI inflows average 4.9% of GDP but are highly dispersed (SD 6.2), ranging from -16.2 to 32.6 . Transactional variables (trade, FDI) and monetary aggregates exhibit substantial cross-country variation. Coverage varies: broad money is observed in 240 country-years, while lending rates are available in only 192 (27% missing).

Table 1.2: Descriptive statistics and data coverage for the processed panel (2000–2023)

Variable	N	Mean	SD	Min	Max	Missing
FDI (% GDP)	260	4.918	6.170	-16.205	32.596	1.5%
Broad money growth (annual %)	240	14.310	11.819	-11.930	61.836	9.1%
Deposit interest rate (%)	213	4.080	3.560	0.122	15.502	19.3%
Real interest rate (%)	205	4.858	8.252	-30.123	41.294	22.3%
Lending interest rate (%)	192	9.783	6.392	3.060	32.000	27.3%
Inflation, GDP deflator (%)	264	5.298	8.235	-21.739	59.080	0.0%
Trade (% GDP)	240	131.923	85.767	32.972	437.327	9.1%
Log GDP per capita	264	24.719	2.019	19.721	27.947	0.0%
Exchange-rate depreciation (%)	253	2.700	30.527	-13.362	476.795	4.2%
Tourism arrivals (Log)	240	15.258	1.150	12.938	17.502	9.1%
Human capital index	240	2.383	0.476	1.516	3.399	9.1%

N = country-year observations. The panel is unbalanced (max 264, min 192 per variable). FDI is expressed as a percentage of GDP. Monetary proxies have the narrowest coverage (lending rate: 192 obs). See the country-level audit tables for structural gaps.

1.4.2 Univariate Distributions

1.4.3 Bivariate Relationships

1.4.4 Correlations and Data Coverage

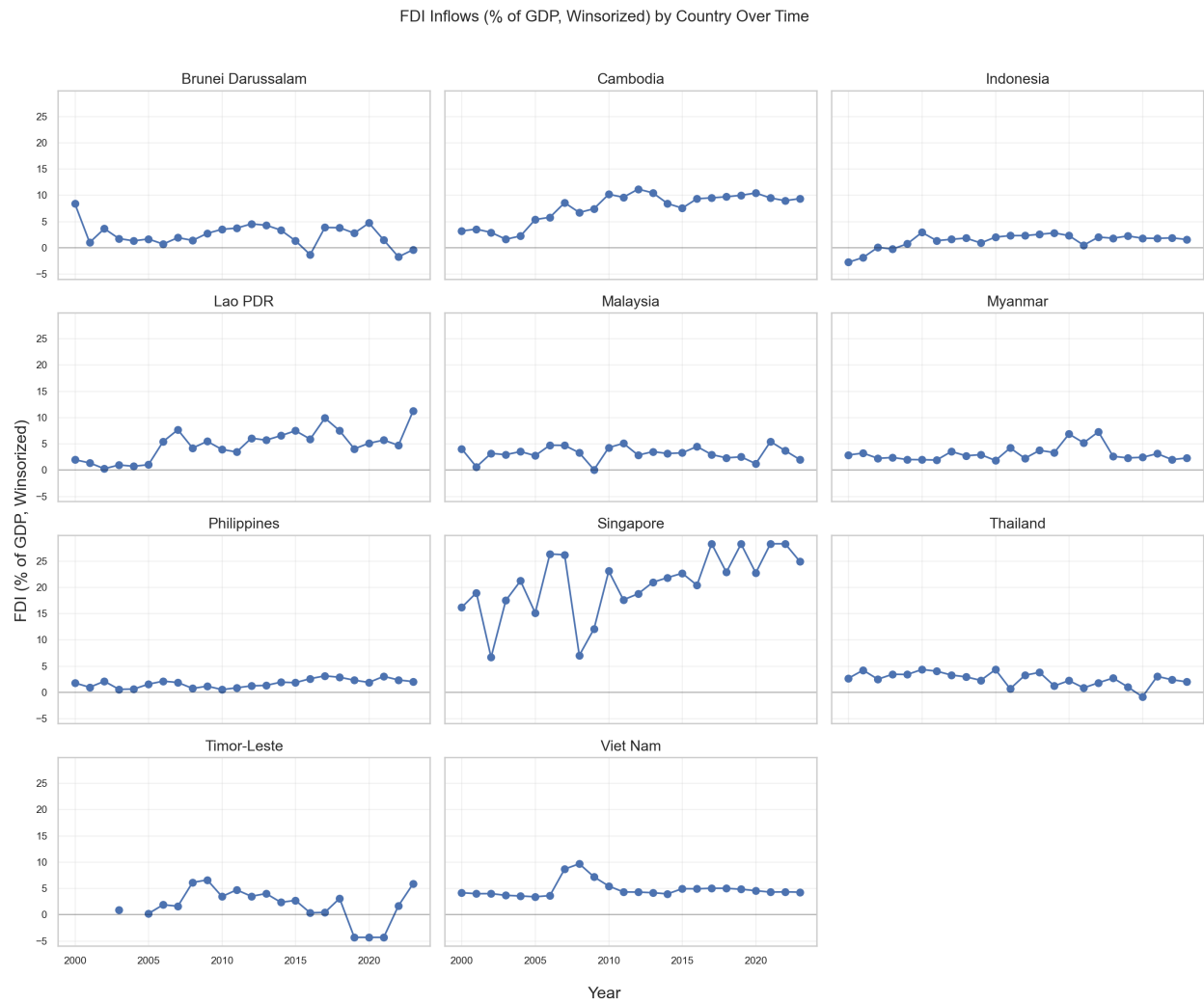


Figure 1.1: FDI net inflows (% of GDP) by country, 2000–2023. Country-level time series reveal substantial heterogeneity in levels and trends across ASEAN-oriented economies. Singapore’s structurally high FDI-to-GDP ratio reflects its role as a regional financial hub.

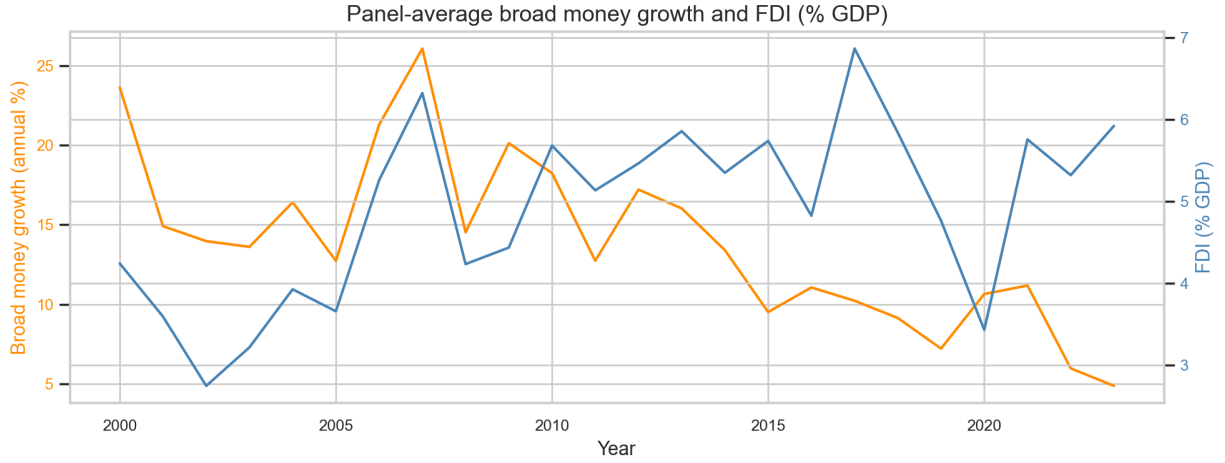


Figure 1.2: Yearly cross-country average of broad money growth and FDI net inflows. Both series show a co-movement pattern consistent with a liquidity channel, particularly during the post-GFC period (2010–2015).

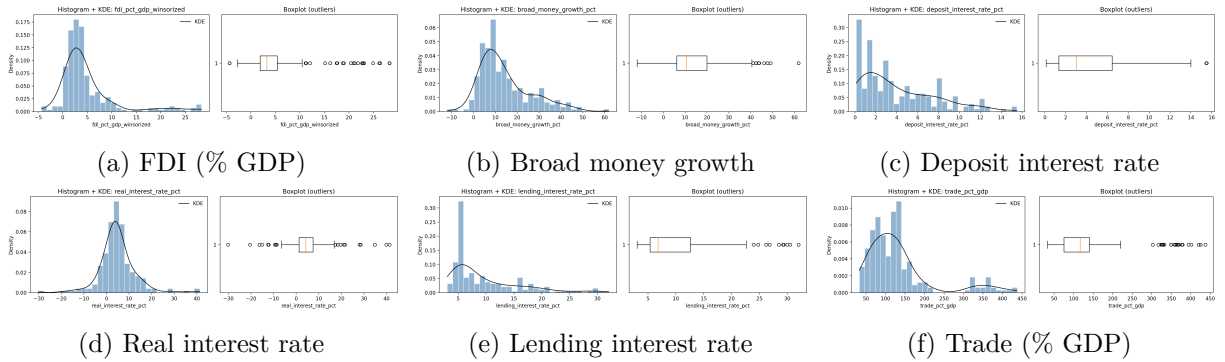


Figure 1.3: Distribution of key variables in the analytical panel. The monetary proxies (broad money, interest rates) and macro controls exhibit right-skewness and substantial dispersion, motivating the use of robust covariance estimation.

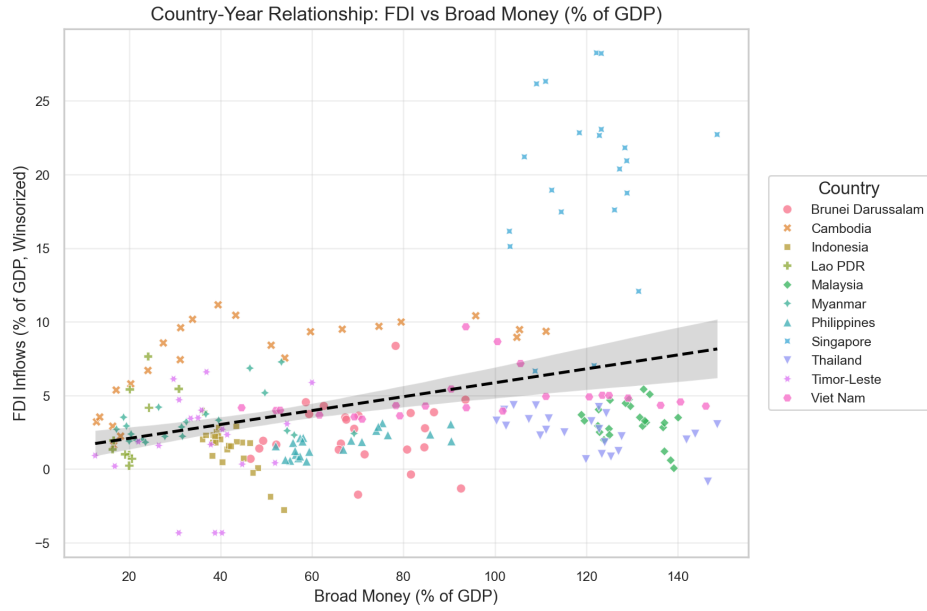


Figure 1.4: Broad money growth vs. FDI net inflows by country-year. The unconditional relationship is flat, consistent with the weak FE-DK coefficients in the main regression table.

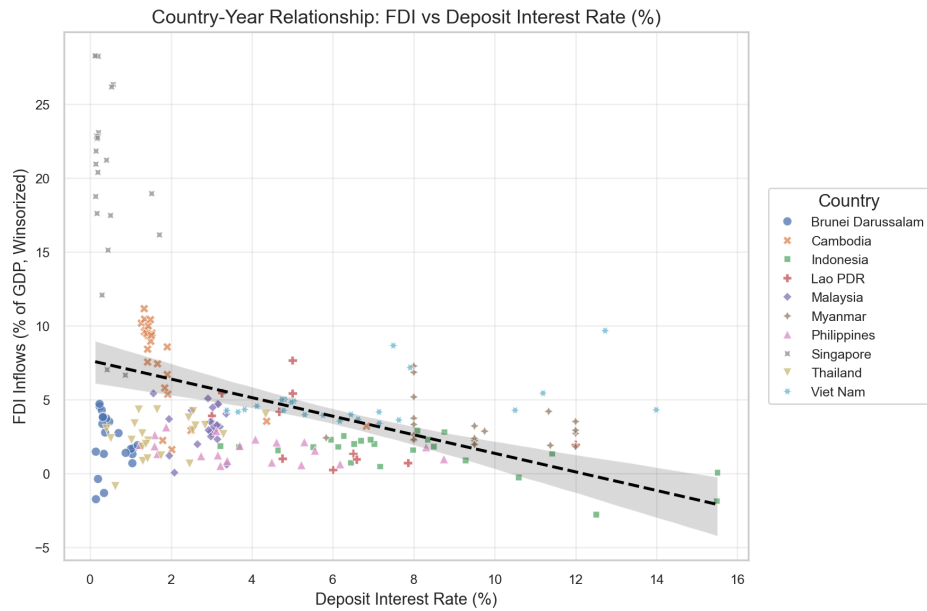


Figure 1.5: Deposit interest rate vs. FDI net inflows. A slight negative slope is visible in the raw data, but the association is overwhelmed by between-country heterogeneity once country fixed effects are included.

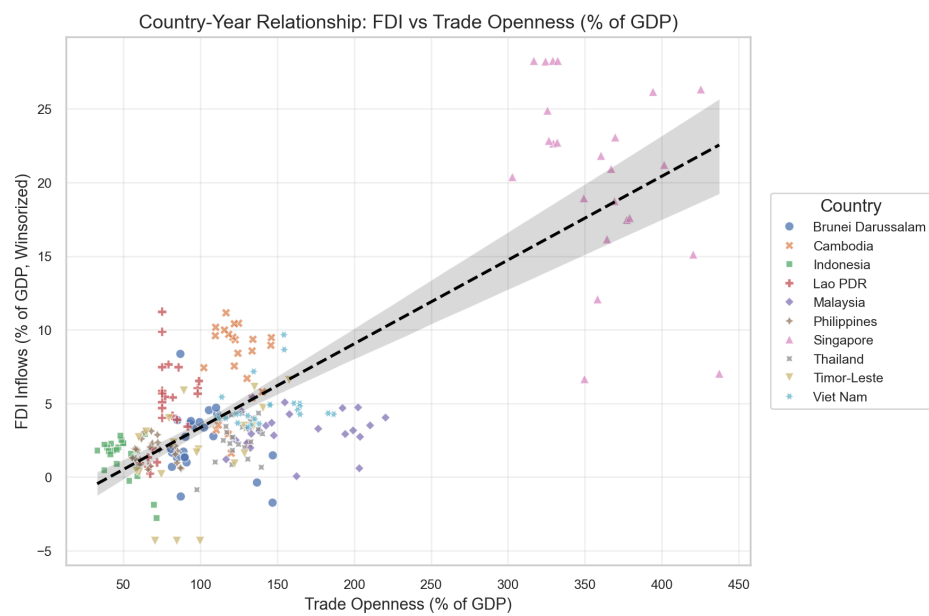


Figure 1.6: Trade openness vs. FDI net inflows. The strong positive unconditional correlation ($r = 0.78$) is driven largely by Singapore; the within-country signal is weaker, consistent with the imprecise trade coefficients in the FE-DK regressions.

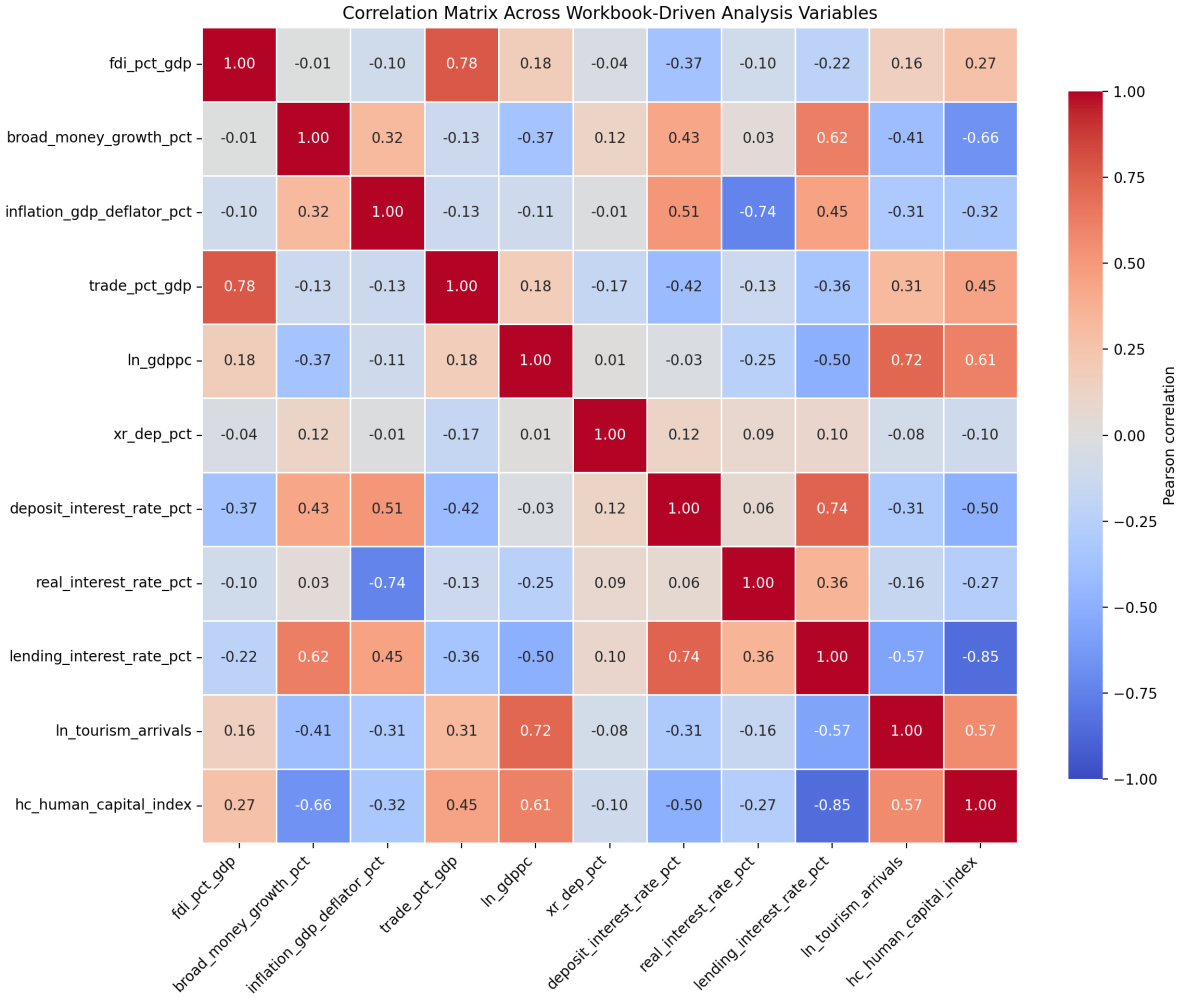


Figure 1.7: Pearson correlation matrix of main analysis variables. FDI is positively correlated with trade openness (0.78) and GDP per capita, and negatively correlated with deposit (-0.37), real (-0.10), and lending (-0.22) interest rates. Broad money growth correlates positively with inflation (0.32) and negatively with the human capital index (-0.67).

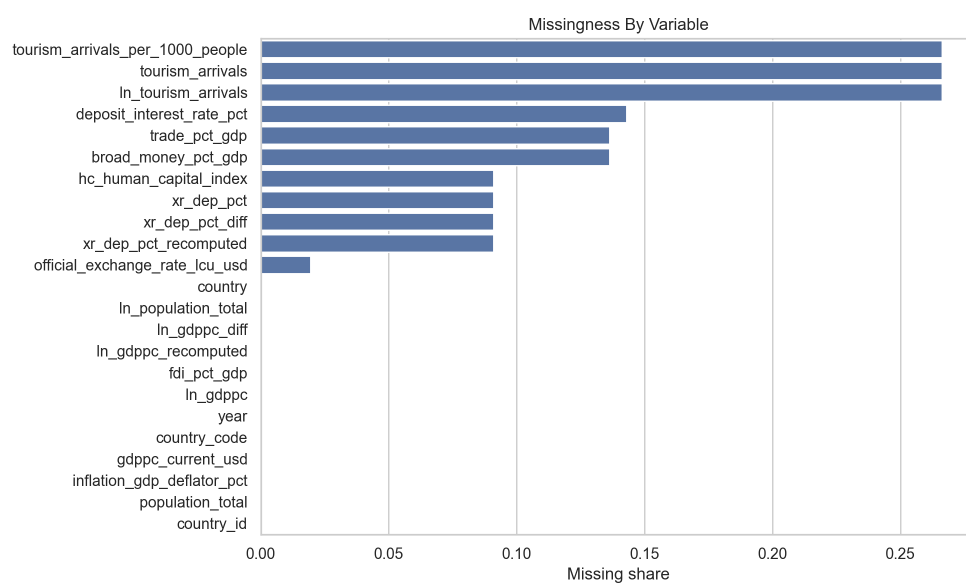


Figure 1.8: Missing data rates by variable. Interest-rate proxies have the highest missingness (19–27%), concentrated in Cambodia, Lao PDR, and Timor-Leste. FDI and GDP per capita are fully observed. Controls have 0–9% missingness.